

**Exoplanet Visualization Using With
Flask**

*Submitted in partial fulfillment of the
requirements for the degree*

of

Bachelor of Science in Electrical and Electronics Engineering

by

Melih Alparslan AKSOY 150717003

Alper ARSLAN 150717013

Under the supervision of

Assoc. Do. Dr. Fulya CALLIALP



**MARMARA
UNIVERSITY**

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

FACULTY OF ENGINEERING

MARMARA UNIVERSITY

January, 2024

DECLARATION OF AUTHORSHIP

Project Title Exoplanet Visualization Using With Flask
Authors Melih Alparslan AKSOY, Alper ARSLAN

Supervisor Assoc. Doc. Dr. Fulya CALLIALP

We, Melih Alparslan AKSOY, Alper ARSLAN, hereby declare that this project titled Exoplanet Visualization Using With Flask is entirely our own work, unless otherwise referenced or acknowledged. The content of this project is the result of our own research and efforts, and we have complied with all ethical guidelines and academic standards.

We are aware of the consequences of academic dishonesty and understand that any violation of ethical standards in this project may lead to disciplinary actions as defined by Marmara University's policies.

Melih Alparslan AKSOY
150717003

Alper ARSLAN
150717013

Istanbul
January, 2024



MARMARA
UNIVERSITY

Electrical and Electronics Engineering
Engineering
Marmara University
Istanbul, Türkiye

Exoplanet Visualization Using With Flask

by

Melih Alparslan AKSOY, Alper ARSLAN

SUBMITTED TO THE DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
DEGREE OF
BACHELOR OF SCIENCE AT
MARMARA UNIVERSITY

The author(s) hereby grant(s) to Marmara University permission to reproduce and to distribute publicly paper and electronic copies of this document in whole or in part and declare that the prepared document does not in any way include copying of previous work on the subject or the use of ideas, concepts, words, or structures regarding the subject without appropriate acknowledgement of the source material.

Signature of Author(s)

Supervisor
Assoc. Prof. FirstName LastName

Head of Department
Prof. FirstName LastName

ACKNOWLEDGEMENTS

We would like to extend our sincere thanks to our supervisor Fulya CALLIALP

We would like to thank our Engineering department where all equipments have been supplied.

We were lucky to meet many great ideas and thoughts about our topic of project.

Finally we would like to thank our family and friends who supported us for this big journey

Melih Alparslan Aksoy, Alper ARSLAN

Marmara University

Date: January, 2024

ABSTRACT

Write the abstract here.

Keywords: Artificial Intelligence, Internet of Things (IoT). (write in alphabet order)

ABBREVIATIONS

AWGN	Additive White Gaussian Noise
BPSK	Binary Phase Shift Keying
MMSE	Minimum Mean Square Error

Contents

1 INTRODUCTION	1
1.1 Thesis Background	1
1.2 Project Definition	1
1.2.1 Problem Statement	2
1.2.2 Context and Scope	2
1.2.3 Significance and Implications	2
1.3 A meaningless section	Error! Bookmark not defined.
2 RESEARCH OBJECTIVE	4
2.1 Adding mathematical expressions	4
2.2 Project Specifications	4
2.2.1 Adding Tables in L ^A TEX:	4
2.3 Another meaningless section	4
2.3.1 A very important subsection	5
3 RELATED LITERATURE	7
4 SYSTEM DESIGN	8
4.1 Realistic constraints and conditions	8
4.2 Cost of the Design	10
4.3 Engineering Standards	10
4.3.1 Bluetooth standards	11
4.3.2 Wireless standards	11
4.3.3 Microprocessor standards	11
4.3.4 Software standards	Error! Bookmark not defined.
4.3.5 Important note	Error! Bookmark not defined.
4.4 Details of the System Design	11
5 METHODS	13
6 RESULTS AND DISCUSSION	19
6.1 Simulation Results	19

7 CONCLUSION	20
7.1 Conclusion	Error! Bookmark not defined.
7.2 Future Recommendation	Error! Bookmark not defined.
A Appendix	20
A.1 A Proof	Error! Bookmark not defined.
A.2 Some Matlab Code.....	21
A.3 An Algorithm	Error! Bookmark not defined.
A.4 Some C code.....	Error! Bookmark not defined.
REFERENCES	23

List of Figures

1.1	A magnificent bird.	2
5.1	A Printed Circuit Board.	10

List of Tables

2.1	Example Table	5
2.2	Your caption.	5

Chapter 1

INTRODUCTION

1.1 Thesis Background

The investigation of the issue this project is trying to solve starts with a thorough analysis of previous findings, hypotheses, and research, with a primary focus on intricate topics like space exploration, astronomy technologies, and public involvement. The purpose of the literature review is to place the significance of the James Webb Space Telescope (JWST) photographs in the larger framework of astronomy communication and study.

1.1.1 Literature Review

The literature study makes use of an extensive knowledge base to provide light on the historical background, current advancements in space travel, and significance of telescopes like the JWST in helping to unravel the secrets of the cosmos. Furthermore, JWST is investigating October's current platforms for distribution of astronomical photos, emphasizing the requirement for a unique approach to dynamically and interactively display its images.

1.1.2 Relevance and Importance

JWST is becoming a technological wonder of the contemporary era, offering unprecedented insights into the farthest reaches of the cosmos. This initiative is significant because it attempts to close the gap between the most recent findings in astronomy and the general public's availability. The intended online application is intended to appeal to a broad spectrum of users, including astronomy hobbyists, educators, and members of the general public, with the goal of democratizing access to JWST photos. This initiative has the potential to have an influence that goes beyond the confines of science and inspires wonder and interest in people all over the world.

1.1.3 Current State and Limitations

While astronomical photographs are presently accessible through platforms, there is still a limited level of interest in JWST images. Real-time updates, community-focused approaches, and interactivity are often absent from current solutions. With the creation of an intuitive, dynamic, and captivating platform for JWST photo research and sharing, this project seeks to do away with these constraints.

1.1.4 Transition

A thorough examination of the project's goals and definition, as well as a transition to an awareness of its suitability and existing constraints, are made possible by the synthesis of the literature. This background section lays the groundwork for a thorough understanding of the problem's context, which makes the transition to next sections easier.

1.2 Project Definition

Define the problem in terms of the technical details and features that the product, service, or process should have. These details are often set by your advisors to make sure the project is challenging and appropriate for a project. Here's how to break it down:

1.2.1 Problem Statement

The primary goal of this project is to develop an intuitive online application that will showcase JWST photos. It will also solve the limitations of current platforms and include interactive features and real-time updates.

1.2.2 Context and Scope

The creation of the web application, the incorporation of real-time updates from the JWST database, and the deployment of interactive features are all included in the problem's scope. Its goal is to establish a forum where individuals may examine, exchange, and converse about JWST photographs.

1.2.3 Significance and Implications

It is crucial to find a solution to this issue in order to encourage public involvement in space exploration. The online application's inability to close this gap may have a number of

unintended repercussions, including the deconstruction of the democratization of astronomical knowledge and the lost chance to introduce the glories of astronomy to a larger audience.

Chapter 2

RESEARCH OBJECTIVE

The primary goals of our project are to create a dynamic online application for interactive image sharing and to disseminate the important visual data that JWST has provided to a large audience.

2.1 Goal: Creating a Web Application that Updates JWST Images in Real-Time

Our project's primary objective is to develop a web application that refreshes the JWST photos in real time, in keeping with this aim. This online application will periodically pull the most recent information from the JWST database, presenting viewers with engaging and current graphics every month.

2.1.1 Measurability: Instant Image Presentation Capacity

The ability of the web application to promptly and flawlessly provide JWST images to each user will be used to determine how measurable this aim is. An evaluation of the application's speed at processing a user's request for a picture and providing the most recent, high-quality JWST image is part of the measurement.

2.1.2 Achievability: Application Development Process

The design of the project's development process in accordance with the team's experience and available technologies will determine whether or not this aim can be achieved. Depending on the team's abilities, the development stages that call for proficiency in a variety of areas, including database administration, programming languages, and user interface design, will be chosen. To accomplish the aims, each step will be carefully coordinated and carried out in accordance with a clearly defined strategy.

2.2 Project Specifications

2.2.1 Technical Requirements

Web Application

- It needs to work with the most recent versions of Google Chrome, Mozilla Firefox, and Safari.
- Responsive design: It ought to adjust to various screen sizes, including desktop, tablet, and mobile.
- Interface that is easy to use: improved user interface for simple JWST image sharing and discovery.

Database

- JWST data should be stored in a scalable, secure database.

Security

- Encrypted user data must be sent and kept.
- Authentication and authorization systems need to adhere to security guidelines.

2.2.2 Performance Expectations

High Speed and Update Frequency

- Real-time JWST picture updates should be possible with this web application.
- It is important to guarantee quick loading and response times.

Database Performance:

- Quick responses to database requests are expected.

2.2.3 Design Elements

Appearance and User Interface

- Users should be able to efficiently view JWST photos thanks to visual design.
- Interactive features and menus should make it easier for consumers to comprehend the information.

2.2.4 Functionalities

Image Discovery

- JWST images ought to be searchable, filterable, and explorable by users.
- Interactive elements such as rotation and zooming ought to be offered.

User Engagement

- Users ought to be able to share their own observations and remark on photos.

2.2.5 Compatibility

Multi-Platform Support

- The online application ought to function flawlessly across several operating systems, including Windows, MacOS, and Linux.

Multiple Browser Compatibility:

- The online program ought to work with a variety of browsers (Google Chrome, Mozilla Firefox, Safari, Microsoft Edge).

Chapter 3

RELATED LITERATURE

Hubble Space Telescope (April 24, 1990, 12:33:51)

Hubble Space Telescope is one of the most important telescopes on work. This is the only telescope designed to be repaired, used again and renovated by astronauts. It is still using over 30 years ,and NASA plans to use it until 2030-2040. Its main goal is to analyze, search and gather information from other planets, and helping scientist to understand universe through galaxies. It mostly use in Milky way. With helping information from Hubble, dark matter and dark energy has been studied. Hubble can be said as the pioneer of upcoming space telescopes because scientist have gained and learned so much thanks to it.

Spitzer Space Telescope (25 August 2003, 05:35:39)

Spitzer Space Telescope is another launching of Space Telescopes. Like Hubble and Webb, it also plays such a huge role to grab space data. It was using with Infrared Wave Spectrum which helps to visualize the exoplanet studying. It worked on dust clouds, star forming and star explosions. However it has been shut down in January 30,2020.

CONTRIBUTION OF THIS STUDY

Launching of these Telescopes helped scientist and astronauts to demonstrate and give new ideas about space dealing. In life when some take risks, it can bring two things: success or experience. If these two launches could have been failed, it brings this one life advice. However, they were launched perfectly and used well. After getting information from Hubble and Spitzer Telescopes mostly as infrared, scientists have blown their broaden and searched on them. It made the Webb Telescope better design, better software and better motion.

Chapter 4

SYSTEM DESIGN

4.1 Realistic constraints and conditions

Space is still unknown and special place for human-beings. It is not stable and therefore it expands taking into consideration of Albert Einstein's idea. As an addition, people are curious and creative creatures. They want to explore and taste different. After remembering the Planet Earth where people live on is in Space, it brings more desire to learn about it. NASA and its cooperatives sent and in time will send satellites, devices, telescopes and so on to accomplish this phenomenon. As it has been mentioned, space is not still and related to time in most cases. When it is the case, humans want to give some answers about most enigmas like 'How did the universe happen?', 'Is time travelling real?', 'Is there any living organisms outer than Earth or any livable planets?' etc.

Therefore, James Webb Space Telescope has been sent by National Aeronautics and Space Administration (NASA), European Space Agency (ESA) and Canada Space Agency (CSA) in 25 December 2021. The conditions were clearer than Hubble Space Telescope because of more experience and information about space, its surroundings, orbital changes etc. These were the conditions which must be thought on.

In the web design, the photograph proofs have been declared to criticize and make an illustration to users. As can be seen, the web design has been designed into three sections: Logo, Navigation bar and home screen. It will be mentioned on Methods section, but it may be mutual for readers and programmer to explain briefly the design of web site and its working condition. In this design Python Flask, CSS and HTML programs have been used and coded by.

There would be another selection like Python Flet instead of HTML and CSS, but the reason to use these programs to make the web site and clearer and more understandable.

The condition parts will be taken apart into 2 parts: Web page design conditions and James Webb Space Telescope (JWST) itself conditions. It also divided into 2 parts : inner and outsider conditions.

4.1.1 James Webb Space Telescope (JWST)

4.1.1.1 Inner Conditions

Sunshield

The sunshield play great role for space crafts to keep them cold. There is a giant star people called it as sun which emits big amount of heat and light. Without sunshield, crafts can be burned or not functioned. The sunshield of James Webb Space Telescope has designed into 5 layers. Their material is called as Kapton. The overall sunshield is much as size of a tennis court.

Optics

The is crucial for space telescopes. They are launched to produce pictures or at least wavelength of space's ingredients. It must be well designed, combined and attached to each other. JWST has been equipped with a mirror which has 6.5(21.3 feet) meters in diameter and 18 hexagonal segments. The telescope has been told to work such wavelengths that others will not access and work on it.

Power Situation

Power situation is important because when it does not work correctly, the space craft would die. So, its layers should be calculated recently and well designed.

JWST has been equipped with solar panels. They are toward to sun, and able to collect sun light and get energy from.

Software

Scientist have clarified JSWT equipment with high technology software and high hardware design. It includes control of instruments, data control, information stack and communication.

4.1.1.2 Outsider Conditions

When a spacecraft handle its mission, outside can affect it. In theory, you may not calculate everything 100% correct so there might be issues when it comes to practice. When taking into account that it works outside of Earth, It is kind of same logic but different aspect because there are some cases even everything work ordinary, some meteor can explode and hit the spacecraft or orbital changes may happen. So this features can be thought as well.

4.1.2 Web Page Design Conditions

Software Design

As programmers and users know that when a software or simply a piece of code has been published, its design is really important to outside world. Designer must attract other people to interact with code or software to use and make it vulnerable. So, as you now think, if it brings attention, it will bring power and fame. It must be looking good for the first sight, must be attracting, must be easy to use, must be simple and not confusing, and must be protectable.

Programming Language

Language is also so crucial to implement and invert it. There are specific programming languages for individual usage. Programmer can decide what he/she wants to use for the aim of idea. Preferably, HTML, CSS and Python Flask has been used for this web page. The reason is that they are fast, trustworthy, easy to implement.

4.2 Cost of the Design

This project is based on software. Therefore, there are no other external objects that we can take account. There can be Internet cost because the web page will work when it is online. Having domain is also another cost for the design. Internet costs around 20 \$ and 40 \$ in Turkey. There is minimum system necessities for computers to compute and evaluate the prediction and person who wants to try and work on this software must have minimally Windows 10 software, good RAM, high memory space, well storage capacity and automation. This kind of computer costs 500 \$ to 1000 \$ less or more. Python-FLASK and HTML-CSS applications are free to use means not extra costs from them.

Name of Component	Its Costs
Internet	10-30 \$
Computer	500 – 1000 \$
Domain	15 \$
Hosting	45 \$

4.3 Engineering Standards

4.3.1 Software Engineering Standards

In this project, Python Language is used because the main topic of this project is to show and explain how important is to explore and understand space. Therefore, Python, HTML and CSS is being used. HTML and CSS are easy and way better to implement web pages and design them. When a python project has been launched, it's sufficient libraries must be used to work with desired functions. Python libraries such as FLASK or Flet are so brief and accessible. When thinking about Python is still improving and developing, it is important to get it. Reader can see with Flet library publishing. Another example is HTML and CSS. Programmer can make a entire whole web page with that two languages. Their features are similar to Python.

This project needs to work in software means well and sufficient GPU has requested. Online clouding system calls as Google Collaboratory takes place here because Google Collaboratory lets programmer to use GPU. To perform and print images and codes GPU has been needed.

4.3.2 Metadata

Data Exchange Metadata helps to use different type of files into different type of program languages. So, in design, HTML and CSS codes have been used inside Python file. Also some jpg and .png extensions have been inside HTML file to print them onto web page.

4.4 Details of the System Design

In this section, details of design will be mentioned clearly and briefly.

The space is always a topic for programmers of this project, and indeed it said that the topic for the Engineering Project shall be space. It was searched about what to do and what to design to contribute to science and decided to make research about James Webb Telescope.

As latest news and information web sites, there are photos coming from James Webb Space Telescope. It was a great opportunity to programmers to program and code this kind of project to combine all James Webb Space Telescope's images.

Then the languages of project have been decided. It would be Python, HTML and CSS. After that to implement and make HTML code running inside Flask library has been found and used.

Chapter 5

METHODS

As seen in below, this is a CSS code to build the background of home page. Background image has been put and set. Its properties have been declared like size, position and height. Then Logo of website has been put to the right side of home page with 20 pixels space to the left.

```
body {
    margin: 0;
    padding: 0;
    background-image: url('https://www.ejiltalk.org/wp-content/uploads/2021/03/pexels-photo-110854.jpeg');
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
    height: 100vh; /* Set to the height you want */
}
.tool{
display:flex;
justify-content:space-between;
}
.logo{
margin-top:10px;
margin-left:20px;
color:#b0dfd1;
font-size:30px;
}
```

Then navigation bar is placed to top right side with helping of FLASK. It can be seen in below picture. The elements of navigation bar(navbar) are located as Home, Photos, Contact and About us. Photos are divided into three parts. Image of 2024, Image of 2023 and Image of 2022 with helping of href in HTML library of a.

```

<body>
  <div class="tool">
    <div class="logo">James Webb Space Telescope News</div>
    <ul>
      <li><a href="{{url_for('home')}}">Home</a></li>
      <li>
        <a href="{{url_for('photo')}}">Photos &#10534;</a>
        <ul class="dropdown">
          <li><a href="{{url_for('p24')}}">2024 Photos</a></li>
          <li><a href="{{url_for('p23')}}">2023 Photos</a></li>
          <li><a href="{{url_for('p22')}}">2022 Photos</a></li>
        </ul>
      </li>
      <li><a href="{{url_for('contact')}}">Contact</a></li>
      <li><a href="{{url_for('about')}}">About us</a></li>
    </ul>
  </div>

```

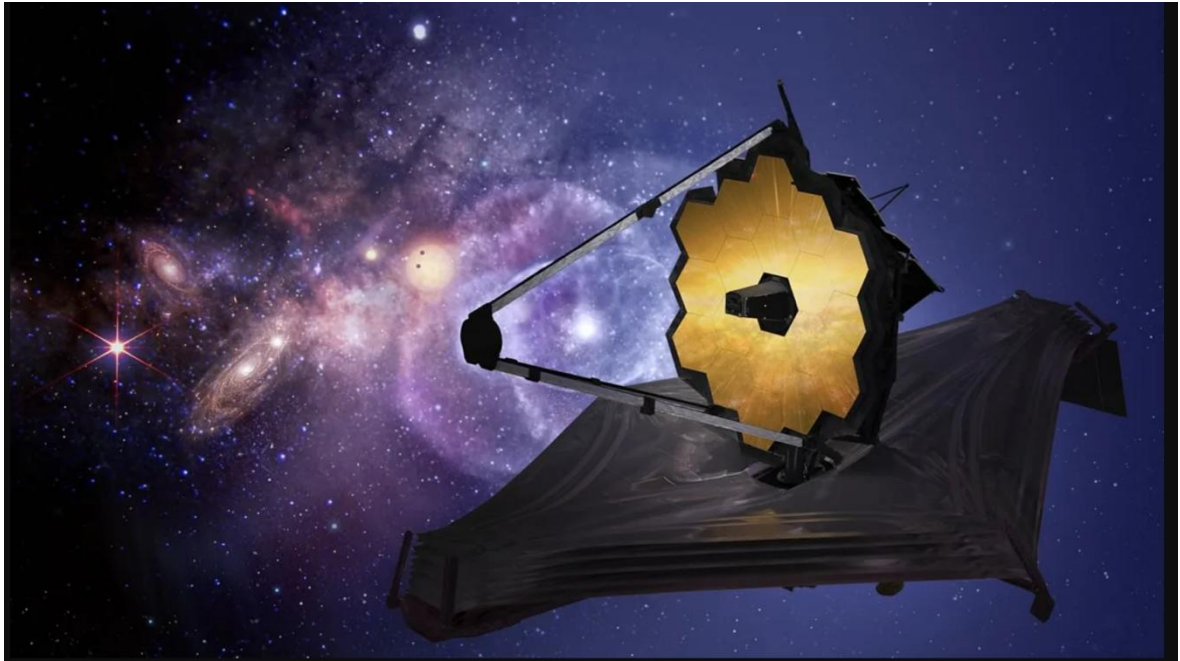
Then on center of web page, two grids have been put to introduce James E. Webb and James Webb Space Telescope as the code seen below.

```

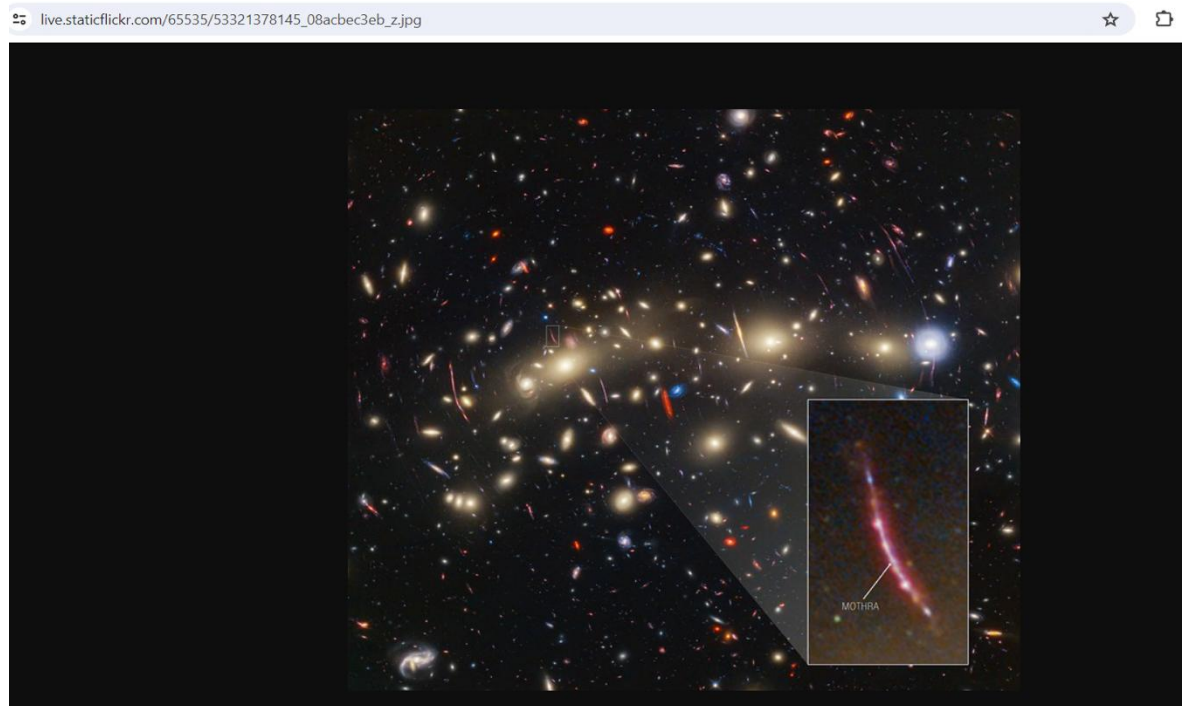
<div class="inf4" style="..">
<a href="https://webb.nasa.gov/" target="blank" style="..">
<div class="inf2">What is the James Webb Space Telescope?</div>
<div class="inf3">
  
  <div style="..">The James Webb Space Telescope (JWST) is a cutting-edge infrared space observatory set to succeed the Hubble Space Telescope. Launching in late

```

When you copy source code and put into web browser, you will get that this image of JWST , and when user clicks to image or its information brackets, it will direct user to James Webb Nasa Web Page.



When a user clicks one of the options of photo selection in navigation bar, it will direct it to specific choice of user. When a user or programmer wants to see all images individually, he/she can copy .jpg or .png link inside href and copy into web browsers and also can see as below.



When picture options happened and user scrolled to the bottom of the page, he/she can see page number menu. It also designed by HTML and CSS. You can see the code below.



This is the CSS code to design HTML command.

```
.pag{
text-align:center;
margin-right:20px;
}
.pag a {
display: inline-block;
padding: 8px 16px;
text-decoration: none;
color: black;
background-color: #ddd;
margin: 0 4px;
}

.pag a.active {
background-color: #333;
color: white;
}
```

This is the HTML command of CSS code.

```

<div class="pag">
    <a href="{{url_for('new1')}}" class="active">1</a>
    <a href="{{url_for('new2')}}">2</a>
    <a href="{{url_for('new3')}}">3</a>
    <a href="{{url_for('new4')}}">4</a>
    <a href="{{url_for('new5')}}">5</a>
    <a href="{{url_for('new6')}}">6</a>
    <a href="{{url_for('new7')}}">7</a>
</div>

```

Python FLASK code has been given below to demonstrate the working principle.

```

@main.route('/')
def funccome():
    return render_template('den.html')
@main.route('/home')
def home():
    return render_template('den.html')
@main.route('/photo')
def photo():
    return render_template('new 1.html')
@main.route('/contact')
def contact():
    return render_template('learning.html')
@main.route('/about')
def about():

```

Google Cloud My First Project Search (/) for resources, docs, products, and more Search

Compute Engine VM instances CREATE INSTANCE IMPORT VM REFRESH LEARN

Virtual machines VM instances Instance templates Sole-tenant nodes Machine images TPUs Committed use discounts Reservations Migrate to Virtual Machin...

INSTANCES OBSERVABILITY INSTANCE SCHEDULES

Instance "ubuntu-web-server" is underutilized. You can save an estimated \$13 per month by switching to the machine type: e2-small. [Learn more](#) DISMISS

VM instances

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	jwnasa-com-web-server	us-west4-b			10.182.0.6 (nic0)	34.125.209.75 (nic0)	SSH
☒	ubuntu-web-server	northamerica-northeast2-a	Save \$13 / mo Save \$26 / mo		10.188.0.3 (nic0)	34.130.193.98 (nic0)	SSH

Related actions HIDE

Web Application kodunu yazdıktan sonra bu uygulamayı bir sunucuda çalıştırmamız ve halka yayınlamamız gerekiyordu ve bunu yapmak için Google Clouds'un sunucularını kullandık. İlk önce Google Clouds'tan bir sunucu satın aldık ve o sunucuya

Chapter 6

RESULTS AND DISCUSSION

Finally, readers can see that this type of scientific experiments will help human-beings to make to or try to make them to understand entire universe, its behavior, time in space and so on. Eventually, this topic might be related to religious conflicts, time travelling after some information gathered. This website can be the pioneer to expand or attract more people to play with space experiments and get more experiences. It can be seen as an example that same pictures getting from JWST are clearer in terms of pixels than Hubble Space Telescope. It means with trying humans can broaden their knowledges and abilities.

This idea can lead to other specific live organisms outside of Earth and people can attract them like in the fictional space movies. There can be found livable planets and people can travel there to investigate and even live on it. Illnesses like cancer, Alzheimer can be cured after researching alien organisms. People can live more and healthier.

Chapter 7

CONCLUSION

Space is extremely big and getting bigger every time. Earth may not be the only planet whose occurs life on it. Humans must keep trying to give more and best to understand life better. Such experiments like launching satellites, sending space shuttles, space stations and space tools will make humans better in terms of knowledges about outer life. There can be good or bad results after all. They may not find something spectacular or reasonable, and all effort can be waste. In another hand, everything may go perfect and find something alive or something different or alien touch. If they want find something and understand time or space better, they shall try.

Appendix A

A.1 Pytgon Code

```
from flask import Flask, render_template

main=Flask(__name__)

@main.route('/')
def funccome():
    return render_template('den.html')
@main.route('/home')
def home():
    return render_template('den.html')
@main.route('/photo')
def photo():
    return render_template('new 1.html')
@main.route('/contact')
def contact():
    return render_template('learning.html')
@main.route('/about')
def about():
    return render_template('learning.html')
@main.route('/new1')
def new1():
    return render_template('new 1.html')
@main.route('/new2')
def new2():
    return render_template('new 2.html')
@main.route('/new3')
def new3():
    return render_template('new 3.html')
@main.route('/new4')
def new4():
    return render_template('new 4.html')
```

```
@main.route('/new5')
def new5():
    return render_template('new 5.html')
@main.route('/new6')
def new6():
    return render_template('new 6.html')
@main.route('/new7')
def new7():
    return render_template('new 7.html')
@main.route('/p23')
def p23():
    return render_template('new 1.html')
@main.route('/p22')
def p22():
    return render_template('new 11.html')
@main.route('/p24')
def p24():
    return render_template('p24.html')
@main.route('/n11')
def n11():
    return render_template('new 11.html')
@main.route('/n12')
def n12():
    return render_template('new 12.html')
@main.route('/n13')
def n13():
    return render_template('new 13.html')
@main.route('/n14')
def n14():
    return render_template('new 14.html')
@main.route('/n15')
def n15():
    return render_template('new 15.html')
```

REFERENCES

[1] <https://webb.nasa.gov/index.html>

[2] <https://www.esa.int/>

[3] <https://www.asc-csa.gc.ca/eng/>

[4] https://en.wikipedia.org/wiki/James_Webb_Space_Telescope

[5] https://en.wikipedia.org/wiki/James_E._Webb

[6] <https://mailchimp.com/de/resources/how-to-buy-a-domain-name/>

[7] <https://www.flickr.com/photos/nasawebbtelescope/albums/72177720305127361/>